

Assessment Task 3: Loan Repayment Classification

32130 - Fundamentals of Data Analytics

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1. Description of the Data Mining Problem

1.1 Problem Type and Business Context

This assignment addresses a supervised binary classification problem. The objective is to predict whether a borrower will repay a loan using borrower demographic, income, credit-history, loan, and debt-burden attributes. The class attribute is `loan_paid_back`, where the value 1 indicates that the loan was repaid and the value 0 indicates that the loan was not repaid or that the borrower presents a default risk.

The practical business application is credit-risk management. A lender who can identify borrowers likely to default before issuing a loan can allocate capital more efficiently, reduce financial losses, and design targeted risk-mitigation strategies. The classifier is a screening aid rather than an automatic approval rule, and its error trade-off must be considered before operational use.

1.2 Dataset Description

The supplied `trainData.csv` contains 18,000 records and 21 columns. After `id` and `monthly_income` are removed and `credit_utilisation` is created, nineteen predictors remain across four categories:

- Demographic: `age`, `gender`, `marital_status`, `education_level`
- Income and employment: `annual_income`, `monthly_income`, `employment_status`
- Loan characteristics: `loan_amount`, `loan_purpose`, `interest_rate`, `loan_term`, `installment`
- Credit and debt: `credit_score`, `debt_to_income_ratio`, `total_credit_limit`, `current_balance`, `delinquency_history`, `num_of_delinquencies`, `num_of_open_accounts`

Dataset property	Value	Implication
Total records	18,000	Large enough for 70/30 CV split
Total columns	21	1 ID + 19 predictors + 1 target
Target: repaid (1)	14,407 (80.04%)	Majority class
Target: default (0)	3,593 (19.96%)	Minority class — main risk class

Dataset property	Value	Implication
Missing values	0 observed	No row-count change
Duplicate rows	0 detected	All 18,000 rows retained

1.3 Class Imbalance and Evaluation Strategy

The target is imbalanced: 80.04% of records are labelled repaid and 19.96% are default. A classifier that predicts repayment for every record would therefore achieve about 80% accuracy without identifying any risky borrower. Accuracy alone is inadequate for this problem. The comparison uses confusion-matrix measures including precision, recall, class F1 and macro F1, with class 0 treated as the default event. ROC curves and AUC values are retained as supporting threshold-independent evidence.

Stratified sampling preserves the class distribution in the training and holdout partitions. Model selection places most weight on macro F1 and default-class F1 while also considering accuracy, AUC and the business consequences of false default flags and missed defaults.

1.4 Link to Previous Exploratory Analysis

The exploratory analysis conducted in Assignment 2 found that `credit_score` and `debt_to_income_ratio` showed the clearest separation between repaid and default borrowers. Employment status was also associated with repayment outcomes. The classification workflow builds on those findings using the larger 18,000-record training dataset and compares five classifier families.

2. How to Approach the Problem

2.1 Classification Framework

The problem is approached as a supervised binary classification task in KNIME Analytics Platform. The pipeline covers data input and audit, preprocessing and feature engineering, stratified holdout partitioning, cross-validation within the training partition, final model fitting, holdout scoring, and interpretation.

The saved workflow uses a stratified 70/30 split with a fixed seed, producing 12,600 training rows and 5,400 holdout rows. Decision Tree, k-NN, RProp MLP and Gradient Boosted Trees use 10-fold stratified cross-validation, while the polynomial-kernel SVM branch uses 5 folds. The learner configurations are then assessed on the holdout data. Encoding and normalisation are performed separately in the training and holdout branches rather than fitted once on the training data and applied to the holdout; this is a workflow limitation and should be corrected in a future implementation.

2.2 Model Selection Strategy

Five classifier families are implemented: Decision Tree, k-NN, MLP neural network, SVM and Gradient Boosted Trees.

Comparing different algorithmic families makes the final recommendation evidence-based rather than dependent on one modelling assumption. The final classifier is selected using the strongest balance across accuracy, macro F1 and default-class F1. Default precision and recall are interpreted separately because a model can be very reliable when it predicts default while still missing many actual defaults. Interpretability is used as a secondary criterion when performance is similar.

3. Data Pre-Processing and Transformation

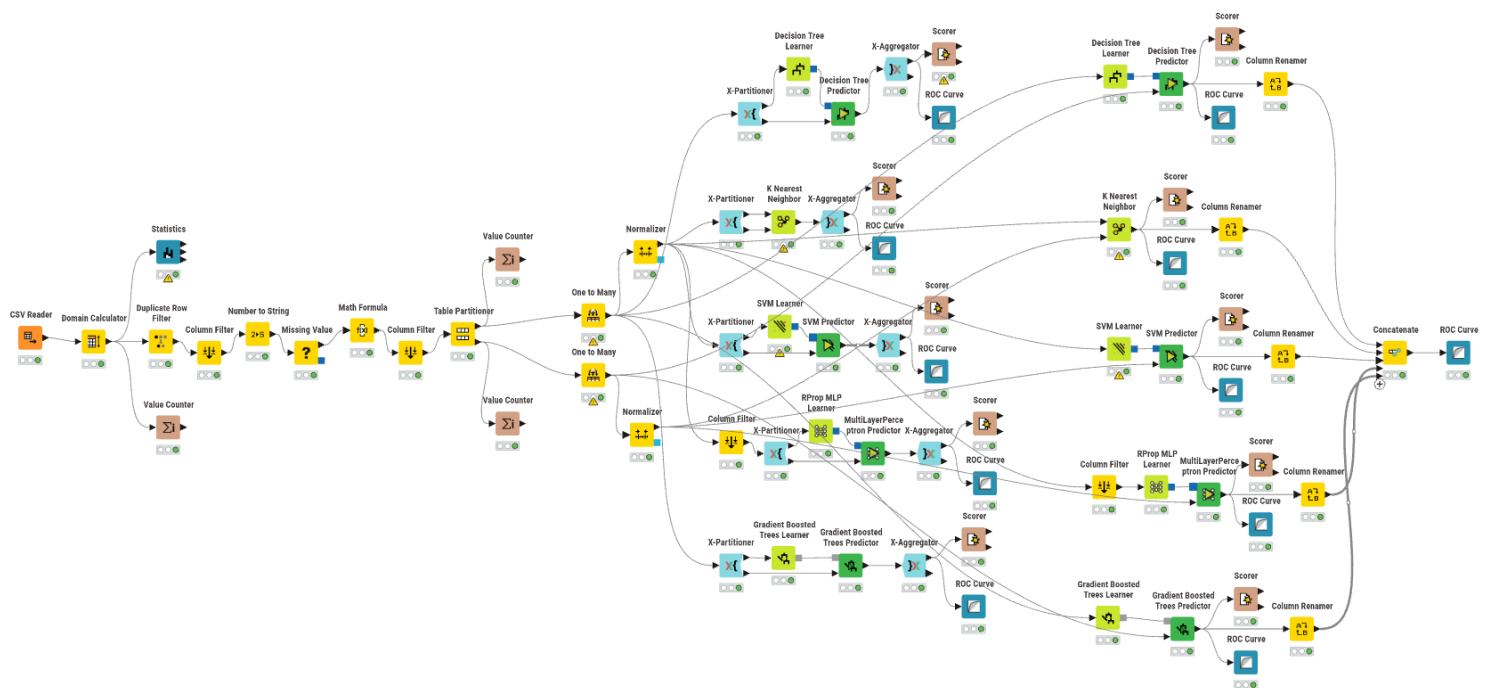


Figure 1: Complete KNIME workflow showing preprocessing, cross-validation, final retraining, and evaluation pipelines for all classifiers.

3.1 Initial Data Audit

The workflow begins with a CSV Reader configured for the supplied comma-delimited file. Domain Calculator, Statistics and Value Counter nodes document the schema, numeric ranges and class distribution. The Duplicate Row Filter retains all 18,000 records, and the Missing Value node also produces no row-count change. The audit therefore shows no removed duplicate rows and no material missing-value replacement in the executed workflow.

3.2 Preprocessing Steps

Step	KNIME node	Configuration	Justification
Load data	CSV Reader	Comma, header, auto type	Imports 18,000-row training file
Recalculate domains	Domain Calculator	All columns	Ensures category values visible downstream
Audit distributions	Statistics + Value Counter	All columns; target count	Documents class imbalance and data quality
Check duplicates	Duplicate Row Filter	Full-row duplicate check	Prevents repeated records biasing training
Remove fields	Column Filters 6 and 10	Exclude id and monthly_income	Removes identifier and redundant income field
Convert target	Number To String	loan_paid_back only	Treats class as nominal for learner nodes
Handle missing values	Missing Value	Fallback configured; no row-count change	Keeps workflow robust to incomplete inputs
Engineer credit ratio	Math Formula	current_balance / total_credit_limit	Creates credit_utilisation
Holdout split	Table Partitioner	70/30 stratified; fixed seed	Preserves class proportions
Encode categoricals	One to Many	Indicator columns appended	Supports numeric learners
Scale numeric features	Normalizer	Z-score on configured columns	Supports distance- and gradient-based learners

3.3 Key Preprocessing Decisions

Removal of id: id is arbitrary and excluded to avoid noise or memorisation. The workflow also removes monthly_income, retains annual_income, and creates credit_utilisation from current_balance and total_credit_limit.

Conversion of the target variable: The loan_paid_back column is originally stored as a numeric integer. KNIME classification learner and Scorer nodes treat class labels more reliably when the target is stored as a nominal string. Number To String is applied exclusively to loan_paid_back. Numeric predictor columns are not converted.

Missing Value node: The node is retained as a configured fallback. The executed workflow shows no observed row-count change from missing-value handling, so no material replacement is claimed.

Categorical encoding: One to Many encoding appends binary indicator columns for nominal predictors. In the exported workflow, several original string columns remain alongside the indicators. KNIME records that k-NN ignores nonnumeric columns and the SVM learner rejects five incompatible columns. A cleaner comparison would filter the original strings consistently so every learner receives the same predictor set.

Scaling: Tree-based classifiers do not require normalisation. k-NN, SVM and MLP use Z-score normalisation because their distance or optimisation behaviour is sensitive to scale. The saved workflow runs encoding and normalisation separately on the training and holdout branches. Future work should fit these transformations on training data and apply the fitted transformations to the holdout data.

Feature engineering: credit_utilisation is calculated as current_balance divided by total_credit_limit. This ratio captures how much of a borrower's available credit is already in use and adds a standard credit-risk indicator not directly available in the raw dataset.

4. Classification Techniques Used - Summary of Parameters Used and Settings

4.1 Cross-Validation Methodology

Cross-validation is applied inside the 70% training partition using X-Partitioner and X-Aggregator nodes. Decision Tree, k-NN, RProp MLP and Gradient Boosted Trees use 10 stratified folds; the polynomial-kernel SVM uses 5 stratified folds. The exported workflow preserves one configuration for each learner and the aggregate validation outputs, but it does not preserve a candidate grid or per-candidate tuning table. The settings below are therefore reported as configured settings rather than as results of a fully documented hyperparameter search.

4.2 Decision Tree

Decision Trees relate directly to the Week 7 material covering impurity, split selection, pruning and confusion-matrix interpretation. The saved learner uses Gini index, no pruning, a minimum of 20 records per node, and average split points. It uses the encoded, unnormalised table because tree-based models are not sensitive to feature scale.

On the 30% holdout set, the Decision Tree achieved 88.87% accuracy, 82.54% default precision, 56.12% default recall, 66.81% default F1, 80.06% macro F1 and 0.863 AUC. Its recall is the highest of the five holdout models, but it generates 128 false default flags, compared with 32 for Gradient Boosted Trees.

The model remains valuable for interpretability because its split structure can be translated into explicit decision rules. It has the highest default recall among the five holdout models, identifying 605 of 1,078 actual defaults. However, 128 repaid applicants are incorrectly flagged as default, four times the 32 false-default flags produced by Gradient Boosted Trees. The tree therefore captures 20 more defaults but creates 96 additional false alarms. A single tree can also be less stable and less accurate than an ensemble because its split structure can change with the training sample.

4.3 K-Nearest Neighbour

k-Nearest Neighbour classifies each record using nearby observations in feature space. The saved learner uses $k = 25$, distance-weighted votes and class-probability output. Z-score normalisation is used because distance calculations are sensitive to feature magnitude. This is the configured setting recorded in the workflow rather than the winner of a preserved candidate grid.

The holdout model achieved 88.22% accuracy, 85.88% default precision, 49.07% default recall, 62.46% default F1, 77.74% macro F1 and 0.823 AUC. It correctly flags 529 defaults, misses 549 defaults and incorrectly flags 87 repaid applicants. The high precision shows that a default prediction is usually reliable, but the 49.07% recall is the weakest default capture among the five models.

Prediction is computationally less efficient on large datasets because distances must be evaluated against many stored observations. The method is also sensitive to irrelevant features, class imbalance and the chosen value of k despite normalisation.

4.4 Support Vector Machine

The saved SVM learner uses a polynomial kernel with $C = 1.0$, $\text{bias} = 1.0$, $\text{power} = 1.0$ and $\text{gamma} = 1.0$. It is therefore described as a polynomial-kernel SVM, not a linear SVM. Z-score normalisation is used on the configured numeric columns. The exported learner also records five incompatible string columns, which should be removed consistently in a revised workflow.

The holdout model achieved 87.85% accuracy, 77.26% default precision, 55.47% default recall, 64.58% default F1, 78.62% macro F1 and 0.764 AUC.

The polynomial SVM correctly classifies 4,146 repayments and 598 defaults. It misses 480 actual defaults and incorrectly flags 176 repaid applicants. These counts produce 77.26% default precision and 55.47% default recall. The model therefore provides moderate default capture, but its 0.764 AUC shows weaker probability ranking than the tree-based and neural-network alternatives. The reported configuration is the saved workflow setting; a comparative kernel-tuning grid is not preserved.

4.5 Multilayer Perceptron Neural Network

The RProp MLP is the Week 9 neural-network classifier. Its saved learner uses one hidden layer with 10 hidden neurons and a maximum of 100 iterations. Random-seed use is disabled in the recorded configuration, and Z-score normalisation supports stable optimisation across differently scaled numeric predictors.

The holdout model achieved 89.57% accuracy, 88.84% default precision, 54.64% default recall, 67.66% default F1, 80.72% macro F1 and 0.890 AUC. It correctly identifies 4,248 repayments and 589 defaults, while missing 489 defaults and incorrectly flagging 74 repaid applicants. These results give it the second-highest macro F1 and AUC in the comparison.

The network captures nonlinear relationships and produces relatively few false default flags. Its main limitation is interpretability: hidden representations cannot be translated into borrower-level decision rules as directly as a tree structure.

4.6 Gradient-Boosted Trees

Gradient Boosted Trees represents the Week 11 boosting ensemble. Trees are added sequentially so that later trees address errors left by the current ensemble. The saved learner is configured with 100 trees, maximum depth 3, information gain ratio, learning rate 0.10, random row sampling and no column sampling. It uses the encoded, unnormalised table.

The holdout model achieved 90.28% accuracy, 94.81% default precision, 54.27% default recall, 69.03% default F1, 81.63% macro F1 and 0.893 AUC. It correctly classifies 4,290 repayments and 585 defaults, misses 493 defaults and incorrectly flags only 32 repaid applicants. It leads the comparison on accuracy, macro F1, default F1 and AUC.

Its high precision means that default predictions are usually correct, but its moderate recall means it misses 493 of 1,078 actual defaults. The model should therefore be described as conservative in predicting default rather than highly sensitive to defaults. It creates few false default flags, but the current operating point is not suitable when the primary objective is to capture nearly all risky borrowers.

4.7 Summary of All Model Results

Model	Configured setting	Accuracy	Def prec	Def recall	Def F1	Macro F1	AUC
Decision Tree	Gini; no pruning; min node 20	88.87%	82.54%	56.12%	66.81%	80.06%	0.863
k-NN	k=25; distance weighted	88.22%	85.88%	49.07%	62.46%	77.74%	0.823
Polynomial SVM	C/bias/power/gamma = 1.0	87.85%	77.26%	55.47%	64.58%	78.62%	0.764
RProp MLP	10 neurons; 1 layer; 100 iterations	89.57%	88.84%	54.64%	67.66%	80.72%	0.890
Gradient Boosted Trees	100 trees, lr 0.1, depth 3	90.28%	94.81%	54.27%	69.03%	81.63%	0.893

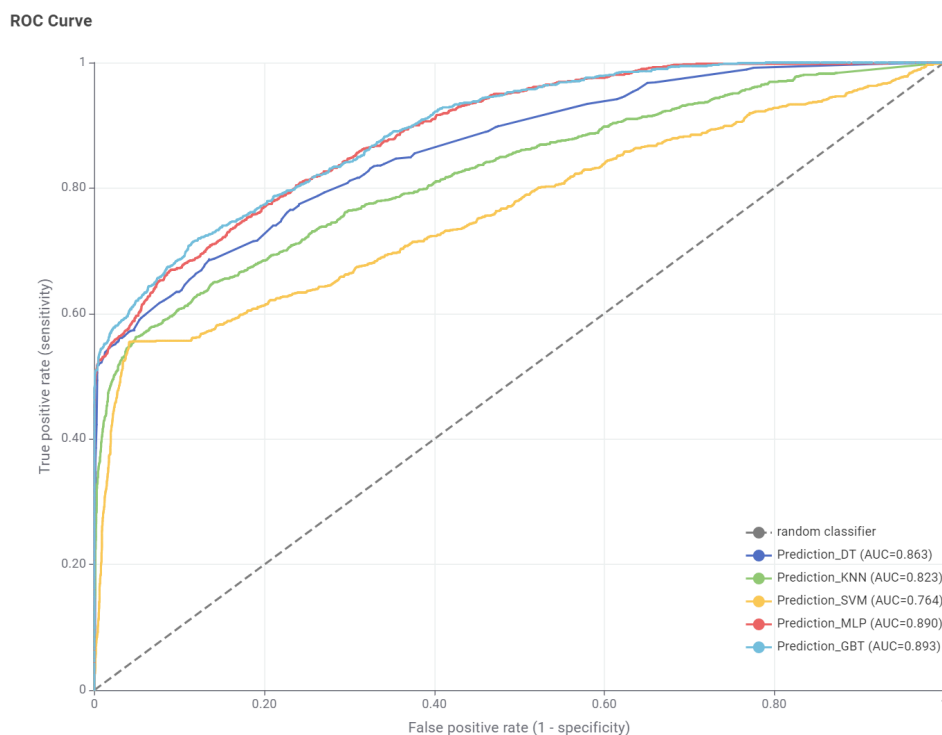


Figure 2: ROC AUC for all models.

The ROC curves show the strongest ranking performance for Gradient Boosted Trees and MLP. The SVM curve remains closer to the diagonal baseline, consistent with its lower AUC. The AUC evidence supports the overall model comparison but does not remove the need to examine the class-specific confusion-matrix results.

5. Justification of Selected Classifier

5.1 Final Model: Gradient Boosted Trees

Gradient Boosted Trees is selected because it leads the holdout comparison on accuracy, default F1, macro F1 and AUC. The model reaches 90.28% accuracy, 94.81% default precision, 54.27% default recall, 69.03% default F1, 81.63% macro F1 and 0.893 AUC.

This result has a specific business interpretation. When the model predicts default, it is correct 94.81% of the time, which limits unnecessary escalation of applicants who would repay. However, it detects only 54.27% of actual defaults. Selection is therefore based on its overall balance and low false-default rate, not on high sensitivity to default risk.

If the lender's main priority is to capture as many defaults as possible, the class threshold or learning objective must be changed. Threshold optimisation, class-weighted learning and explicit misclassification costs should be evaluated before operational deployment.

5.2 Confusion Matrix Analysis

The KNIME Scorer stores predicted classes in rows and actual classes in columns. With default class 0 treated as the positive event, the Gradient Boosted Trees holdout matrix is:

predicted / actual	repaid - 1	default - 0
predicted repaid (1)	4290	493
predicted default (0)	32	585

The model correctly identifies 4,290 repayments and 585 defaults. It incorrectly flags 32 repaid applicants as default and misses 493 defaults by predicting repayment. Default precision is therefore $585 / (585 + 32) = 94.81\%$, while default recall is $585 / (585 + 493) = 54.27\%$.

The model is reliable when it predicts default, but it identifies only slightly more than half of the actual defaults at this operating point.

Holdout TN/FN/FP/TP counts are: Decision Tree 4,194/473/128/605; polynomial SVM 4,146/480/176/598; RProp MLP 4,248/489/74/589; k-NN 4,235/549/87/529; and Gradient Boosted Trees 4,290/493/32/585.

5.3 Comparison with Alternative Models

The Decision Tree has the highest default recall at 56.12%, slightly above Gradient Boosted Trees, but it creates four times as many false default flags and has lower accuracy, macro F1 and AUC. Its transparent split rules remain useful when explanation is the main priority.

k-NN has the lowest default recall and macro F1. Although 85.88% of its default predictions are correct, it misses 549 of 1,078 actual defaults and remains computationally demanding at prediction time.

The polynomial SVM provides moderate recall but the lowest accuracy and AUC. Its lower ranking performance weakens it as an alternative despite a slightly higher default recall than the selected model.

The RProp MLP is the closest alternative, with strong accuracy, macro F1 and AUC. It incorrectly flags 74 repayments, compared with 32 for Gradient Boosted Trees, and its hidden representations are less interpretable.

Overall, Gradient Boosted Trees provides the strongest balance for this workflow. The recommendation remains conditional on the lender accepting a conservative default-prediction threshold that misses 493 defaults in the holdout set.

6. Reflection

6.1 Lessons About Evaluation Metrics

This assignment demonstrates why accuracy alone is unsuitable for imbalanced classification. With about 80% repayments, a majority-only classifier could appear accurate while detecting no defaults. Precision, recall, class F1, macro F1, ROC curves and AUC are needed to describe model behaviour. Accuracy and AUC describe different aspects of performance, while class-specific measures reveal how errors are distributed.

The most important lesson is to state the confusion-matrix orientation and event class before calculating precision and recall. High default precision can coexist with moderate default recall when the model predicts default infrequently. That is the pattern produced by Gradient Boosted Trees in this workflow: its default flags are usually correct, but 493 actual defaults are still missed.

6.2 Importance of Proper Workflow Design

The holdout split keeps model fitting and cross-validation within the training branch. The stratified X-Partitioner and X-Aggregator nodes provide more reliable estimates than a single training split. The workflow also demonstrates that tree-based models and scale-sensitive models require different preprocessing. Tree models use the encoded, unnormalised table, whereas k-NN, SVM and MLP use configured Z-score normalisation.

A limitation is that encoding and normalisation are run separately in the training and holdout branches. This means the holdout transformation is not produced by applying a transformation fitted only on training data. A revised workflow should learn encoding and scaling parameters on the training partition, apply those transformations unchanged to the holdout partition, remove the original string columns consistently, and ensure every learner receives the same predictors. This correction is important because the original report overstated the workflow's leakage protection.

6.3 Comparison of Classifier Families

Decision Trees provide interpretability but lower aggregate performance. k-NN is sensitive to distance relationships and offers the weakest default recall in this comparison. The polynomial SVM provides a clear margin-based method but weaker AUC. MLP captures nonlinear relationships and performs strongly, although its internal representations are difficult to explain. Gradient Boosted Trees achieves the strongest overall holdout results while still allowing feature-importance analysis.

The confusion matrices show that the algorithms reach their scores through different error patterns. Decision Tree captures the most defaults but creates more false alarms; k-NN misses the most defaults; MLP and Gradient Boosted Trees are more conservative in assigning class 0.

No classifier is universally optimal. The best model depends on whether the lender values overall balance, low false-default rates, high default capture, interpretability, or another operational objective.

6.4 Ensemble Learning Insights

Gradient Boosted Trees outperforms the single Decision Tree on accuracy, default precision, default F1, macro F1 and AUC, showing the advantage of sequential boosting for structured tabular data. However, boosting does not automatically solve class imbalance. The selected model remains conservative and captures only slightly more than half of actual defaults at the saved operating point. Its gain over a single tree is therefore strongest in precision and aggregate balance rather than default recall. Threshold selection remains a separate business decision.

The result illustrates the practical trade-off between false alarms and missed risk. Here, only 32 repaid applicants are incorrectly flagged as default, but 493 actual defaults are missed. A real lender would need to set the decision threshold using the relative cost of those errors.

6.5 Future Improvements

Future work should first rebuild the preprocessing pipeline so transformations are fitted on training data and applied unchanged to validation and holdout data. Class-weighted learning, cost-sensitive evaluation or resampling within each cross-validation fold could then be tested to improve minority-class recall without creating an unacceptable number of false default flags.

Threshold optimisation using ROC or precision-recall analysis should be used to select an operating point that matches the lender's risk tolerance. Probability calibration and a fully recorded hyperparameter search would improve reproducibility. Feature-importance or local explanation methods could also support transparent review of individual risk predictions.

Overall, successful data mining depends on careful preprocessing, reproducible validation, correct interpretation of evaluation metrics and a model-selection rule aligned with the business cost of errors.

AI-use acknowledgement. ChatGPT was used to assist with report structuring, explanation drafting and language refinement. The workflow implementation, model outputs and evaluation values were checked against the supplied KNIME workflow evidence.